

What APEX Can Do

A student-facing view of the Semester Planning Agent

Live demo: <https://technion-semester-planner-agent.vercel.app>

APEX is built for the messy part of semester planning: a student has passed some courses, failed others, wants a reasonable workload, may have blocked days, and still needs to make real degree progress. The agent turns that situation into a checked plan and keeps helping when the student changes their mind.

From One Message To A Plan

The student can write naturally in Hebrew or English. APEX extracts the semester, track, pace, passed courses, failed courses, unavailable weekdays, requested courses, grades, and any special constraints. If something important is missing, it asks for that missing detail instead of guessing.

Once it has enough information, APEX builds a plan with course numbers, course names, credits, exam dates, workload, and a short explanation of the tradeoffs. The final plan is checked by deterministic code before it reaches the student.

What The Student Gets

Output	Why it matters
Verified course list	Courses are checked for prerequisites, duplicates, already-passed courses, unknown numbers, credit range, and mandatory progress.
Weekly timetable	Sections are checked so the plan does not hide class-time collisions.
Exam view	Moed A and Moed B dates are considered, with warnings when exams are too close.
Workload and risk	The plan weighs difficulty, review data, heavy-course stacking, and future impact.
Human explanation	The agent explains why this combination was chosen and what, if anything, is still unresolved.

Example: A Verified Plan

In this example, APEX produced a complete 18.5-credit semester with a moderate workload score. The student can immediately see the courses, credits, exams, and verification status.

Semester Plan

הנדסת מערכות מידע

✓ Verified · 18.5 credits · workload 46.8/100

00940224 · 4 pts · exam 2027-02-23

מבני נתונים ואלגוריתמים

00940241 · 3 pts · exam 2027-02-03

ניהול מסדי נתונים

00940424 · 3.5 pts · exam 2027-02-07

סטטיסטיקה 1

00940594 · 3.5 pts · exam 2027-02-17

עקרונות הכלכלה להנדסים

03240521 · 2 pts

יזמות בארגונים-התפתחויות ומגמות

03240675 · 1.5 pts · exam 2027-01-29

ספרדית למתחילים

03940820 · 1 pts

חינוך גופני - תנועה ומחול

Built by my Semester Planning Agent 🤖

A checked semester plan: verified, 18.5 credits, workload 46.8/100.

Course Advice, Not Just Scheduling

APEX can also answer direct questions about courses. For example, if the student asks whether a course is worth retaking to improve a grade, the answer is based on several signals together: the student's grade, the course's historical average, the student's own average, CheeseFork difficulty, satisfaction, and review text.

This matters because the same grade can mean different things for different students. A 70 can be a warning sign for a student whose average is much higher, but less important for another student whose overall average is close to 70.

you

האם אתה ממליץ לי לשפר קורס? לפי הציונים שלי

AGENT

לפי הנתונים, **סטטיסטיקה 1** הוא מועמד סביר לשיפור: קיבלת בו **70**, שזה קצת מתחת למוצע ההיסטורי של **71.8**, אבל הרבה מתחת למוצע האישי הידוע שלך (**86.5**), כך שהציון הזה בולט לרעה ביחס אליך. מצד שני, המשוב על הקורס חזק מאוד — יש **41** חוות דעת, עם שביעות רצון ממוצעת **4.07/5** וקושי **3.49/5**, וביקורות כמו "**מרצה מדהימה באמת**" ו-"**הכי טובה שיש!**" מרמזות שהקורס עצמו סביר ואף אהוד, כך ששיפור יכול להיות כדאי בעיקר אם אתה חושב שתוכל לחזור קרוב יותר לרמתך הרגילה.

backend: <https://technion-semester-palnn>

The agent explains why Statistics 1 may be worth retaking, using grades, averages, and reviews.

Follow-Up Changes

APEX does not treat every follow-up as a brand-new planning problem. If the student asks to add, remove, or swap a course, it keeps as much of the previous plan as possible and changes only what needs to change.

When a request creates a real conflict, the agent does not silently drop the request. It explains the conflict and asks for explicit confirmation before forcing an exception.



Here's your updated plan. I kept 7 of your 8 courses, removed בסטיסטיקה לדוקטורנטים, and added 19.5 credits total.

Still open: class time overlaps with (11:30-13:30) התנהגות ארגונית in every section combination; only 1 mandatory course(s) in the plan, expected at least 3 - a semester should make real degree progress unless the student asked for fewer.

Note: סדרות עתיות וחזיון was removed from the plan because its class times collide with another course in every available section combination - a clashing week is never delivered. On עיבוד שפה טבעית חן: you asked to add it, but I couldn't - its class times collide with ניהול פרויקטים in every available section combination. Want me to add it anyway despite that? Confirm and I will, no matter what.

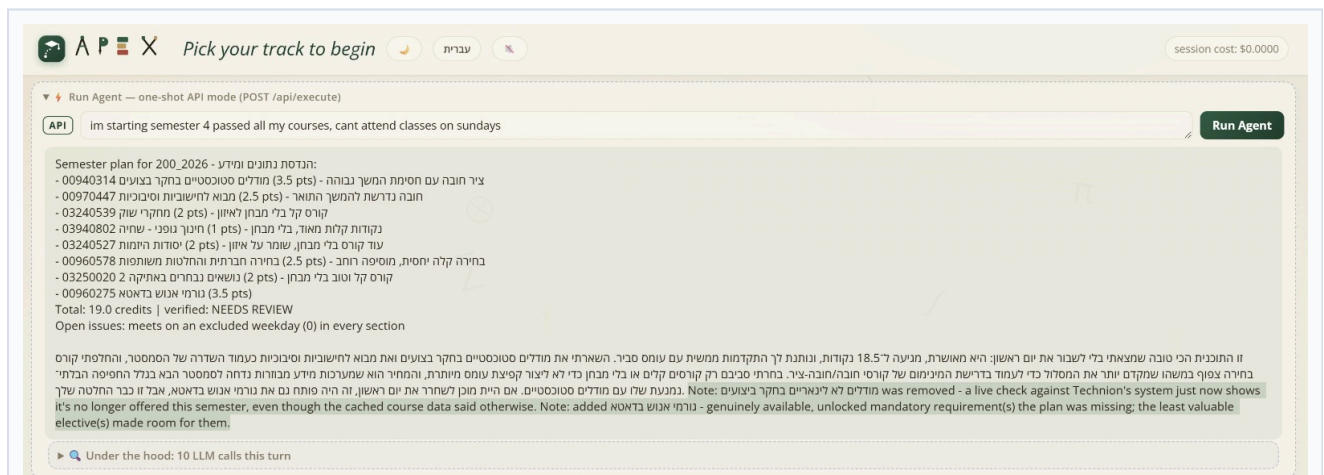
Anything you'd like to change? Pick a quick action or reply below.

A follow-up revision that preserves most of the old plan and clearly names the remaining conflict.

Fresh Offering Check Before Delivery

The bundled catalog is useful for fast planning, but it can become stale during registration season. Before the final answer, APEX runs a live offering check against Technion's current system.

In one live run, the cached data still showed Nonlinear Models as available. The live check found it was no longer offered next semester, so APEX removed it from the plan and filled the missing progress with an available mandatory course instead. This is exactly the kind of quiet failure a planner should catch before the student trusts the result.



Run Agent — one-shot API mode (POST /api/execute)

API: im starting semester 4 passed all my courses, cant attend classes on sundays

Run Agent

Session cost: \$0.0000

Semester plan for 200_2026:

- 00940314 מודלים סטוכסטיים בחקר בצועים (3.5 pts)
- 00970447 מבוא לחשיבויות וסיבוכיות (2.5 pts)
- 03240539 מחקר שוק (2 pts)
- 03940802 קורס קלוד מאוד, בלי מבחן (1 pts)
- 03240527 נקודות קלות מאוד, בלי מבחן (2 pts)
- 00960578 עזר קורס בלי מבחן, שומר על איזון (2 pts)
- 03250020 בחירה קלה יחסית, מוסיפה רוחב (2.5 pts)
- 00960275 קורס קל וטוב בלי מבחן (3.5 pts)

Total: 19.0 credits | verified: NEEDS REVIEW

Open issues: meets on an excluded weekday (0) in every section

ו החלפתי קורס בחירה צפוף במשהו שמתקדם יותר את המסלול כדי לעמוד בדרישת המינימום של קורסי חובה/חובה-ציר. בחרתי סביבם רק קורסים קלים או בלי מבחן כדי לא ליצור קפיצת עומס מיותרת, והמחיר הוא שמערכות חידע מבוזרות נדחה לסמסטר הבא בגלל החסימה הבדתי. Note: מודלים לא לינאריים בחקר בצועים - a live check against Technion's system just now shows it's no longer offered this semester, even though the cached course data said otherwise. Note: נורמי אנוש בדאטא - genuinely available, unlocked mandatory requirement(s) the plan was missing; the least valuable elective(s) made room for them.

Under the hood: 10 LLM calls this turn

Live check example: Nonlinear Models was removed because Technion's current system showed it was no longer offered.

Retakes And Grades

Failed retakes are treated as high priority because they often block future courses. Grade-improvement retakes are handled more carefully: APEX can suggest one, but it does not add it silently. The student must approve it first.

Grades can come from the chat or from an uploaded transcript PDF. The system does not fetch private Technion account data.

What Else It Handles

- Three supported Technion tracks.
- Hebrew and English intake.
- Passed, failed, and explicitly requested courses.
- Blocked weekdays and section-level timetable constraints.
- Light, normal, and fast pace preferences.
- Course reviews from CheeseFork.
- Historical grade averages from technion-histograms.
- Transcript PDF parsing into courses and grades.
- Optional Supabase session restore using a hashed student email.
- Live offering check before final delivery.
- Roadmap, risk report, weekly schedule, and exam study plan tools.
- Required API endpoints with prompt, response, and model-step trace.

Limits And Future Work

APEX is a planning assistant, not a registration system. It does not enroll students, access private Technion accounts, or promise that cached catalog data will update by itself. When Technion publishes a new catalog, the offline pipeline should be rerun.

Future work could add focused retrieval for academic regulations, forms, and administrative guidance. The core course checks should remain deterministic because prerequisites, schedules, credits, and exam dates need exact answers.